

Relaxation and Odd-Even Effect in Fermi Liquids

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Introduction

The 2D electron gas

- Classical conductors are often modeled as an **electron gas cloud**
- Electrons behave like free, classical particles bouncing off a lattice
- **The problem:** Real electrons are quantum mechanical fermions
- Pauli exclusion principle prevents particles from stacking in low-energy states
- **How does electron clouds in conductors relax to equilibrium when disturbed?**

1.1

Boltzmann equation

Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}} \right) f(t, \mathbf{x}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{x}, \mathbf{p})]$$

1

$f(t, \mathbf{x}, \mathbf{p})$: One-particle density function

$\frac{\partial f}{\partial t}$: Local time evolution

$\mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{r}}$: Streaming/advection term

$\mathbf{F} \cdot \frac{\partial f}{\partial \hbar \mathbf{p}}$: Force/acceleration term due to external fields

$\hat{\mathcal{J}}[f]$: Collision integral - effect of collisions on distribution function

1.2

Fermi Dirac distribution

Fermi-Dirac distribution

$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$$

2

- $\beta = 1/k_B T$ is the inverse temperature
- $\varepsilon(\mathbf{p}) = \frac{\hbar^2 \mathbf{p}^2}{2m}$ is the energy of particles with momentum $\mathbf{p}$
- μ is the chemical potential

The Fermi-Dirac distribution is in **equilibrium**.

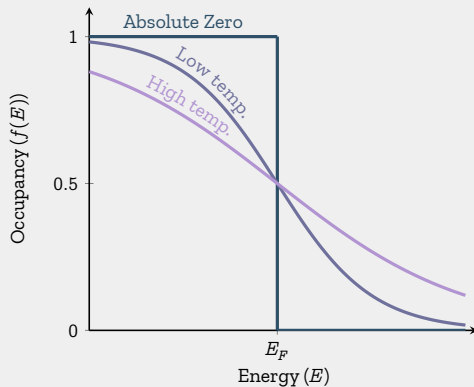


Figure 1: FERMI-DIRAC DISTRIBUTION AT DIFFERENT TEMPERATURES

- At $T = 0$, all states below the Fermi energy are filled, all states above are empty.
- As T increases, the distribution smooths out.

k -space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k -space corresponds to a wave $e^{i\mathbf{k}}$
- **Fourier transform** converts between k -space to real space

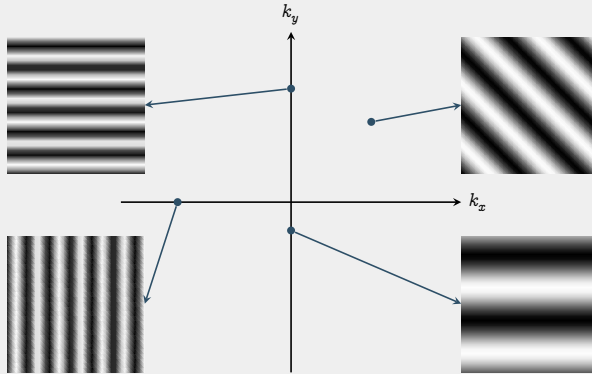


Figure 2: REPRESENTATIVE WAVES OF k -SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k -space $\implies$ Wave direction

Fermi surface

- Surface in k -space that separates *occupied electron states* from *unoccupied electron states*
- Determined by periodicity and symmetry of crystal lattice.
- Changes with temperature.

Collision integral

$$\hat{\mathcal{J}}[f] = - \int \dots \int \frac{1}{(2\pi)^6} d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B}, \quad 3$$

$$\mathcal{B} = f_1 f_2 (1 - f_{1'}) (1 - f_{2'}) - f_{1'} f_{2'} (1 - f_1) (1 - f_2), \quad 4$$

$$f_i = f(\mathbf{p}_i), \quad f_{0,i} = f_0(\mathbf{p}_i). \quad 5$$

$W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2)$: Collision rates

$\mathcal{B}$: "Product of Fermi factors"¹

¹Non-standard in literatures

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}} \right) f(t, \mathbf{x}, \mathbf{p}) = - \int \dots \int \frac{1}{(2\pi)^6} d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B} \quad 6$$

- Very hard (might be impossible) to solve analytically
- Non-linear

Ideas of attacking the equation

- Consider only perturbations near equilibrium: $f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$
- Linearize $\hat{\mathcal{J}}[f]$
- Let there be no external force and no thermal gradient

We will end up with

$$\hat{\mathcal{J}}[\phi] = -\gamma\phi$$

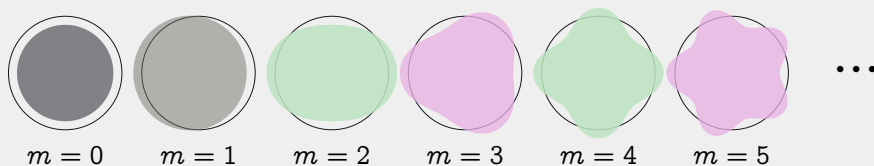


Figure 3: CIRCULAR HARMONICAL PERTURBATIONS OF THE FERMI SURFACE

1. Perturbation is smooth
2. Collision integral is *linear*

The perturbation can be expanded in the harmonics basis: $e^{im\theta}$.

2

Recovering equations

2.1

Linearization of the collision integral

Preparation

Fermi equilibrium:
$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{1}{T} f_0(1 - f_0) \quad 8$$

$$\frac{\partial f_0}{\partial \mu} = \frac{1}{T} f_0(1 - f_0) \quad 9$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{\partial f_0}{\partial \mu} \quad 10$$

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

Perturbation

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$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta \mu$$

Taylor series. Fix ε

12

$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta \mu$$

13

Perturbation of interest: $\delta \mu = T\psi$ (temperature dependent change in chemical potential)

$$\delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} T \psi \quad 14$$

$$= \left(-\frac{1}{T} f_0 (1 - f_0) \right) \times T \psi \quad 15$$

$$= f_0 (1 - f_0) \psi \quad 16$$

Linearization

Taylor expansion of $\mathcal{B}$ at f_0 :

$$\mathcal{B} = \mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}} \quad 17$$

The linear part is:

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1} \right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2} \right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f'_1} \right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f'_2} \right)_{f_0} \delta f_{2'} \quad 18$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} = f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2}), \quad 19$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} = f_{0,1}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1}f_{0,2}(1 - f_{0,1}), \quad 20$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{1'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,2'}) - f_{0,2'}(1 - f_{0,1})(1 - f_{0,2}), \quad 21$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,1'}) - f_{0,1'}(1 - f_{0,1})(1 - f_{0,2}). \quad 22$$

Substitute $\delta f_i = f_{0,i}(1 - f_{0,i})\psi_i$. E.g.,

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = \left[f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2}) \right] \times f_{0,1}(1 - f_{0,1})\psi_1 \quad 23$$

$$= \left[f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})(1 - f_{0,1}) + f_{0,1}f_{0,1'}f_{0,2'}(1 - f_{0,1})(1 - f_{0,2}) \right] \psi_1$$

$$= \left(f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) \right) \left[(1 - f_{0,1}) + f_{0,1} \right] \psi_1 \quad 24$$

$$= \left(f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) \right) \psi_1 \quad 25$$

Define $F_{121'2'} = f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) = f_{0,1'}f_{0,2'}(1 - f_{0,1})(1 - f_{0,2})$. So that

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = F_{121'2'}\psi_1 \quad 26$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_i}\right)_{f_0} \delta f_i = F_{121'2'} \psi_i \quad 27$$

Hence,

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_{1'}}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} \quad 28$$

$$= F_{121'2'} [\psi_1 + \psi_2 - \psi_{1'} - \psi_{2'}] \quad 29$$

$$= F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2)] \quad 30$$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B} \quad 31$$

$$= -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) (\mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}}) \quad 32$$

- Collision integral vanishes at equilibrium: $\hat{\mathcal{J}}[f_0] = 0$
- Throw out $\mathcal{B}_{\text{non-lin}}$

$$\begin{aligned} \hat{\mathcal{J}}[f] = & -\frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}_1 d\mathbf{p}'_2 \\ & \times W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2)] \quad 33 \end{aligned}$$

Expression of $F_{121'2'}$

- For convenience, write $F_{121'2'}$ as **total momentum** (P) and **center of momentum** (Q)

$$\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2, \quad \mathbf{Q} = \frac{1}{2}\mathbf{p}_1 - \frac{1}{2}\mathbf{p}_2.$$

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$$\mathbf{p}_1 = \frac{1}{2}\mathbf{P} - \mathbf{Q}, \quad 35$$

$$\mathbf{p}_2 = \frac{1}{2}\mathbf{P} + \mathbf{Q}, \quad 36$$

$$\mathbf{P} \cdot \mathbf{Q} = \frac{1}{2}(p_1^2 - p_2^2). \quad 37$$

$$\varepsilon(\mathbf{p}_1) = \frac{\hbar^2 p_1^2}{2m^*} = \frac{\hbar^2}{2m^*} \left(\frac{1}{4}P^2 + Q^2 - \mathbf{P} \cdot \mathbf{Q} \right), \quad 38$$

$$\varepsilon(\mathbf{p}_2) = \frac{\hbar^2 p_2^2}{2m^*} = \frac{\hbar^2}{2m^*} \left(\frac{1}{4}P^2 + Q^2 + \mathbf{P} \cdot \mathbf{Q} \right). \quad 39$$

Define²

$$\xi = \frac{\hbar^2 \beta}{2m^*}, \quad \text{and} \quad X = \beta \left(\frac{\hbar^2 P^2}{8m^*} + \frac{\hbar^2 Q^2}{2m^*} - \mu \right) \quad 40$$

Then,

$$\beta(\varepsilon(\mathbf{p}_1) - \mu) = \beta \left(\frac{\hbar^2}{8m^*} P^2 + \frac{\hbar^2}{2m^*} Q^2 - \mu \right) - \frac{\hbar^2 \beta}{2m^*} \mathbf{P} \cdot \mathbf{Q} \quad 41$$

$$= X - \xi \mathbf{P} \cdot \mathbf{Q}. \quad 42$$

Similarly,

$$\beta(\varepsilon(\mathbf{p}_2) - \mu) = X + \xi \mathbf{P} \cdot \mathbf{Q}. \quad 43$$

² X can possibly be interpreted as: *average deviation from background chemical potential*

Hence,

$$f_0(\mathbf{p}_1) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X - \xi \mathbf{P} \cdot \mathbf{Q}]} \quad 44$$

$$f_0(\mathbf{p}_2) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X + \xi \mathbf{P} \cdot \mathbf{Q}]} \quad 45$$

Recall that $F_{121'2'} = f_0(\mathbf{p}_1)f_0(\mathbf{p}_2)(1 - f_0(\mathbf{p}'_1))(1 - f_0(\mathbf{p}'_2))$

$$f_0(\mathbf{p}_1)f_0(\mathbf{p}_2) = \frac{1}{\exp[X - \xi \mathbf{P} \cdot \mathbf{Q}] \exp[X + \xi \mathbf{P} \cdot \mathbf{Q}]} \quad 46$$

$$= \frac{1}{\exp[2X] + \exp[X](\exp[\xi \mathbf{P} \cdot \mathbf{Q}] + \exp[-\xi \mathbf{P} \cdot \mathbf{Q}]) + 1} \quad 47$$

$$= \frac{1}{\exp[2X] + 2 \exp[X] \cosh(\xi \mathbf{P} \cdot \mathbf{Q}) + 1} \quad 48$$

$$= \frac{1}{2(\cosh X + \cosh(\xi \mathbf{P} \cdot \mathbf{Q}))} \quad 49$$

By symmetry,

$$(1 - f_0(\mathbf{p}'_1))(1 - f_0(\mathbf{p}'_2)) = \frac{1}{2(\cosh X + \cosh(\xi \mathbf{P}' \cdot \mathbf{Q}'))} \quad 50$$

$$F_{121'2'} = \frac{1}{4[\cosh X + \cosh(\xi \mathbf{P} \cdot \mathbf{Q})][\cosh X + \cosh(\xi \mathbf{P}' \cdot \mathbf{Q}')] } \quad 51$$

Expression of W - the collision rate

$$W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) = \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \\ \times \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 - \mathbf{p}'_2) \delta(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}'_1) - \varepsilon(\mathbf{p}'_2)) \quad 52$$

$$\hat{\mathcal{J}}[f(t, \mathbf{p}_1)] = - \frac{1}{(2\pi)^6} \int \dots \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 \times \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \\ \times \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 - \mathbf{p}'_2) \delta(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}'_1) - \varepsilon(\mathbf{p}'_2)) \\ \times F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2)] \quad 53$$

Change of coordinates: $\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2$, $\mathbf{Q} = \frac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2)$.

$$\begin{aligned}
 \hat{\mathcal{J}}[f(t, \mathbf{p}_1)] &= -\frac{1}{(2\pi)^3 \hbar} \int \dots \int d\mathbf{p}_2 d\mathbf{P}' d\mathbf{Q}' \delta(\mathbf{P} - \mathbf{P}') \delta\left(\frac{\hbar^2}{2m^*}(p_1^2 + p_2^2 - p_1'^2 - p_2'^2)\right) \\
 &\quad \times |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')] \\
 &= -\frac{1}{(2\pi)^3 \hbar} \int \dots \int d\mathbf{p}_2 d\mathbf{Q}' \delta\left(\frac{\hbar^2}{4m^*}(4Q^2 - 4Q'^2)\right) \\
 &\quad \times |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')] \\
 &= -\frac{1}{(2\pi)^3 \hbar} \int \dots \int d\mathbf{p}_2 Q' dQ' d\Omega \frac{m^*}{\hbar^2} \delta(Q^2 - Q'^2) \\
 &\quad \times |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')] \\
 &= -\frac{m^*}{4\pi \hbar^3} \int \dots \int \frac{d\mathbf{p}_2}{(2\pi)^2} \int d\Omega |V|^2 F_{121'2'} [\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')]
 \end{aligned}$$

2.2

Eigenvalue problem

Reducing the Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}} \right) f(t, \mathbf{r}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{r}, \mathbf{p})] \quad 54$$

- No external force: $\mathbf{F} = 0$,
- No thermal gradient: $\frac{\partial f}{\partial \mathbf{r}} = 0$.

The equation reduces to

$$\frac{\partial f}{\partial t} = \hat{\mathcal{J}}[f] \quad 55$$

Recall that

- $f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$
- $\delta f(t, \mathbf{p}) = f_0(1 - f_0)\psi(t, \mathbf{p})$
- $\frac{\partial f_0}{\partial t} = 0, \quad \hat{\mathcal{J}}[f_0] = 0$

Substituting,

$$\frac{\partial}{\partial t}(f_0(\mathbf{p}) + \delta f(t, \mathbf{p})) = \hat{\mathcal{J}}[f_0(\mathbf{p}) + \delta f(t, \mathbf{p})] \quad 56$$

$$\frac{\partial}{\partial t}f_0(1 - f_0)\psi(t, \mathbf{p}) = \hat{\mathcal{J}}[f_0(1 - f_0)\psi(t, \mathbf{p})] \quad 57$$

$$\frac{\partial \psi(t, \mathbf{p})}{\partial t} = \hat{\mathcal{J}}[\psi(t, \mathbf{p})] \quad 58$$

Proposed ansatz $\psi(t, \mathbf{p}) = e^{-\gamma t} \phi(\mathbf{p})$ (Chemical potential decays with time)

So, the equation reduces to an eigenvalue problem:

$$\hat{\mathcal{J}}[\phi(\mathbf{p})] = -\gamma \phi(\mathbf{p})$$

If *perturbation is smooth*, it can be expanded in the circular harmonics basis:

$$e^{im\theta}, \quad m \in \mathbb{Z} \quad 60$$

For a test mode m ,

$$\hat{\mathcal{J}}[e^{im\theta}] = -\gamma_m e^{im\theta} \quad 61$$

Our goal is to study the eigenvalues γ_m as a function of m and temperature T .

3

Odd-Even effect

Trivial solution

$\gamma_m = 0$ is associated with **conserved quantities**.

- $\gamma_0 = 0$

1. $\phi_0 = 1$: particle number
2. $\phi_0 = \varepsilon(\mathbf{p})$: energy

- $\gamma_1 = 0$

1. $\phi_1 = e^{i\theta}$: x -momentum
2. $\phi_1 = e^{-i\theta}$: y -momentum

Near the Fermi surface,

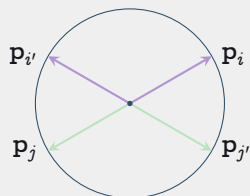
$$\varepsilon(\mathbf{p}_1) \approx \varepsilon(\mathbf{p}_2) \approx \varepsilon(\mathbf{p}'_1) \approx \varepsilon(\mathbf{p}'_2) \approx \varepsilon_F. \quad 62$$

Thus, **conservation of energy** implies **conservation of momentum magnitude**:

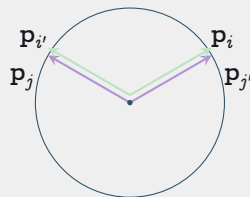
$$p_1 + p_2 = p'_1 + p'_2. \quad 63$$

Three types of collisions that satisfy this constraint:

- Head-on collisions: $\mathbf{p}_1 = -\mathbf{p}_2, \mathbf{p}'_1 = -\mathbf{p}'_2$
- Forward exchange: $\mathbf{p}'_1 = \mathbf{p}_2, \mathbf{p}'_2 = \mathbf{p}_1$
- No collision: $\mathbf{p}'_1 = \mathbf{p}_1, \mathbf{p}'_2 = \mathbf{p}_2$



Head-on process



Exchange process

Figure 4: RELEVANT COLLISION TYPES

No collisions: Has no contribution to the collision integral.

Forward exchange & No collisions: • $\mathbf{p}_1 = \mathbf{p}'_1, \mathbf{p}_2 = \mathbf{p}'_2$.

- $\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2) = 0^3$
- $\therefore$ These collisions has no contribution to the collision integral.

Only **head-on collision** contribute to the collision integral.

³Recall that this is a multiplier in the collision integral

Head on collision

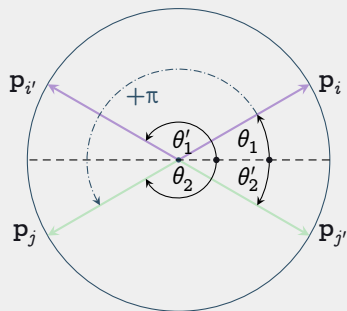


Figure 5: HEAD-ON COLLISION
ANGLE

$$\theta_2 = \theta_1 + \pi, \quad \text{and} \quad \theta'_2 = \theta'_1 + \pi. \quad 64$$

Substituting test mode m , and recalling $e^{im(\theta+\pi)} = (-1)^m e^{im\theta}$, then

$$\begin{aligned} & \psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}'_1) - \psi(\mathbf{p}'_2) \\ &= (1 + (-1)^m) e^{im\theta_1} - (1 + (-1)^m) e^{im\theta_{1'}}. \end{aligned} \quad 65$$

m is odd: the term vanishes; thus the collision integral vanishes.

m is even: the term does not vanish.

Finding the Even Decay Rate : Overview

- Use **auxiliary variable** $\delta_1 = X - \xi \mathbf{P} \cdot \mathbf{Q}$ and $\delta_2 = X + \xi \mathbf{P} \cdot \mathbf{Q}$, together with $\delta_1 = 0$, and $\theta_{12} = \theta_2 - \theta_1$.
- Collision integral over $\mathbf{p}_2$ and Ω . Decompose $\mathbf{p}_2$ into **magnitude** p_2 and **angle** θ_2 .
- **Change integration variable** from p_2 to δ_2 and θ_2 to $\theta_{12} = \theta_2 - \theta_1$.
- **Omega integral**: Approximate far from $\theta_{12} = 0, \pi$, reduces to standard hyperbolic integral. Gains a T factor.
- **δ_2 integral**: Standard numeric integral. Gains another T factor from change of variables.
- **θ_{12} integral**: Seemingly diverges, but cut off by Ω integral. Gains a $\log(T)$ factor.

Finding the Even Decay Rate : Auxiliary Variables

Recall $X - \xi \mathbf{P} \cdot \mathbf{Q} = \beta(\varepsilon(\mathbf{p}_1) - \mu)$, $X + \xi \mathbf{P} \cdot \mathbf{Q} = \beta(\varepsilon(\mathbf{p}_2) - \mu)$. Define

$$\delta_1 = X - \xi \mathbf{P} \cdot \mathbf{Q}, \quad \text{and} \quad \delta_2 = X + \xi \mathbf{P} \cdot \mathbf{Q}. \quad 66$$

$$\implies X = \frac{\delta_1 + \delta_2}{2}, \quad \text{and} \quad \xi \mathbf{P} \cdot \mathbf{Q} = \frac{\delta_2 - \delta_1}{2} \quad 67$$

Assuming $\varepsilon(\mathbf{p}_1) \approx \mu$, $\delta_1 \approx 0$, so $X \approx \xi \mathbf{P} \cdot \mathbf{Q} \approx \frac{\delta_2}{2}$.

Finding the Even Decay Rate : Auxiliary Variables

Define $\theta_{12} = \theta_2 - \theta_1$. Then

$$P = 2p_F |\cos(\theta_{12}/2)|, \quad \text{and} \quad Q = p_F |\sin(\theta_{12}/2)|. \quad 68$$

$$\implies \xi \mathbf{P} \cdot \mathbf{Q}' = \xi P Q \sin \Omega = \xi p_F^2 |\sin(\theta_{12})| \sin \Omega. \quad 69$$

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Let $\Lambda = \varepsilon_F \beta = \xi p_F^2$. Finally, $\xi \mathbf{P} \cdot \mathbf{Q}' = \Lambda |\sin(\theta_{12})| \sin \Omega$.

Finding the Even Decay Rate : Integration Variables

The collision integral has the form (note the first (...) contains constants, the second contains dimensionless integration quantities like ϕ).

$$(\dots) \int \dots \int d\mathbf{p}_2 d\Omega F_{121'2'}(\dots) \quad 71$$

Using $d\mathbf{p}_2 = p_2 dp_2 d\theta_2$, $d\theta_2 = d\theta_{12}$, and $d\delta_2 = \beta d\varepsilon(p_2) = \beta \frac{\hbar^2 p_2}{m^*}$, we obtain the form

$$(\dots) \int \dots \int d\delta_2 d\theta_{12} d\Omega F_{121'2'} \cdot (\dots) \quad 72$$

with constants absorbed into the first (...) (restored later on). Recall $\beta = \frac{1}{k_B T}$. **This is our first T factor.**

Finding the Even Decay Rate : $F_{121'2'}$

$$\begin{aligned}
 F_{121'2'} &= \frac{1}{4[\cosh X + \cosh(\xi \mathbf{P} \cdot \mathbf{Q})][\cosh X + \cosh(\xi \mathbf{P} \cdot \mathbf{Q}')] } & 73 \\
 &= \frac{1}{4[\cosh(\delta_1 + \delta_2)/2 + \cosh(\delta_2 - \delta_1)/2][\cosh(\delta_1 + \delta_2)/2 + \cosh(\Lambda \sin \Omega |\sin(\theta_{12})|)] } \\
 &= \frac{1}{4[\cosh \delta_2/2 + \cosh \delta_2/2][\cosh \delta_2/2 + \cosh(\Lambda |\sin(\theta_{12})| \sin \Omega)] } & 74 \\
 &= \frac{1}{8 \cosh \delta_2/2 [\cosh \delta_2/2 + \cosh(\Lambda |\sin(\theta_{12})| \sin \Omega)] } & 75
 \end{aligned}$$

where we have used $\delta_1 = 0$.

Finding the Even Decay Rate : **Omega Integral**

Collecting terms that depend on Ω , we have

$$\int_0^{2\pi} \frac{d\Omega}{\cosh \delta_2/2 + \cosh (\Lambda |\sin(\theta_{12})| \sin \Omega)}. \quad 76$$

For θ_{12} not near 0 or π , approximate $\sin \Omega \approx u$, extend the integral to $(-\infty, \infty)$, and change integration variables to $v = \Lambda |\sin(\theta_{12})| u$. Then the integral becomes

$$= \int_{-\infty}^{\infty} \frac{d\Omega}{\cosh \delta_2/2 + \cosh (\Lambda |\sin(\theta_{12})| \Omega)} = \frac{1}{\Lambda |\sin(\theta_{12})|} \int_{-\infty}^{\infty} \frac{dv}{\cosh \delta_2/2 + \cosh v} \quad 77$$

$$= \frac{2\delta_2}{\Lambda |\sin(\theta_{12})| \sinh \delta_2/2} \quad 78$$

Where we have used the standard integral $\int_{-\infty}^{\infty} \frac{dx}{\cosh a + \cosh x} = \frac{2a}{\sinh a}$. (**Lambda is our second T factor**).

Finding the Even Decay Rate: δ_{12} Integral

Collecting terms that depend on δ_2 , we have

$$\frac{1}{8 \cosh \delta_2/2} \frac{2\delta_2}{\Lambda \sin(\theta_{12}) |\sinh \delta_2/2|} = \frac{\delta_2}{2\Lambda \sin(\theta_{12}) |\sinh \delta_2|}. \quad 79$$

Using the standard integral $\int_{-\infty}^{\infty} \frac{x}{\sinh x} dx = \frac{\pi^2}{2}$, we have

$$\int_{-\infty}^{\infty} \frac{\delta_2}{2\Lambda \sin(\theta_{12}) |\sinh \delta_2|} d\delta_2 = \frac{\pi^2}{4\Lambda |\sin(\theta_{12})|}. \quad 80$$

Finding the Even Decay Rate : θ_{12} Integral

Collecting terms that depend on θ_{12} , we have

$$\int_0^{2\pi} \frac{d\theta_{12}}{\Lambda |\sin(\theta_{12})|}. \quad 81$$

The integral is divergent at 0 and π . However, it is cut off by the Ω integral. The cutoff occurs when $\Lambda |\sin(\theta_{12})| \sim 1 \implies |\theta_{12}| \sim T/T_F$. Elsewhere, the antiderivative equals $\ln(\tan(\theta_{12}/2)) + C$.

For θ_{12} at least T/T_F from 0 or π , we have

$$= \int_{T/T_F}^{\pi-T/T_F} \frac{d\theta_{12}}{\Lambda |\sin(\theta_{12})|}. \quad 82$$

$$= \frac{1}{\Lambda} [\ln(\tan((\pi - T/T_F)/2)) - \ln(\tan(T/T_F/2))] \quad 83$$

$$= \frac{1}{\Lambda} [-2 \ln(\tan(T/T_F/2))] \quad 84$$

Using small angle approximation, $\tan(T/T_F/2) \approx T/T_F/2$, we have

$$\int_0^{2\pi} \frac{d\theta_{12}}{\Lambda |\sin(\theta_{12})|} \approx \frac{2}{\Lambda} \ln(T_F/T). \quad 85$$

Finding the Even Decay Rate : Final Result

Collecting all the T factors, we have

- One from β in the change of variables from p_2 to δ_2 .
- One from the Λ from Ω integral.
- One log factor from the θ_{12} integral.

Combining constants obtained during the change of integration variable, we obtain

Collision Integral Approximation to Zero-th order

$$-\gamma_m \phi = \hat{\mathcal{J}}[\phi] = -\frac{k_B \pi^2 m^{*2} |V|^2}{\hbar^4 (2\pi)^3} \frac{T^2}{T_F} \ln\left(\frac{T_F}{T}\right) \phi \quad 86$$

Thus, we have $\gamma_m \propto T^2 \log T$. Note that this only works for small ϕ .

Strategy for the Odd Decay Rate

Since, to the zeroth order, the ϕ terms vanish, we need the second order perturbation (first order also vanishes). The strategy is to

- Develop eigenvalue perturbation around $\delta_T = T/T_F \ll 1$
- Expand the ϕ around 0 to obtain a power series in $\varepsilon(p_i)/\varepsilon_F = \delta_T u_i$, where $u_i = \varepsilon_i/T$.
- Integrate the power series against the Fermi functions.
- Use the eigenvalue perturbation theory to obtain the lowest order approximation of γ_m .

4

Further works

van Hove singularity

van Hove singularity: An **energy level** that has infinitely many available states.^a

^aSpike in the density of states

- More available states $\implies$ stronger pairing interactions (forming Cooper pairs)
- Superconductivity can happen

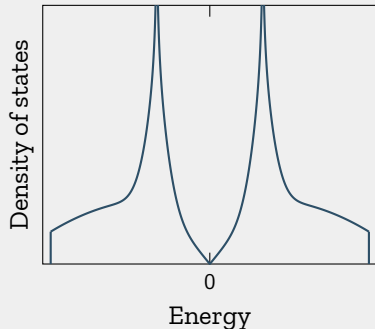


Figure 6: VAN-HOVE SINGULARITY
(NOT TO SCALE, NOT REAL DATA)

If Fermi surface has the **energy that's close to the van Hove singularity**, superconductivity can occur more easily.

- Does the odd-even effect persists near the van Hove singularity?
- What's the relaxation time?

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Thank you!